



Exercise1: INTRODUCTION

1. Introduction to Python libraries for Data Mining: NumPy, Pandas, Matplotlib etc.

Write a Python program to do the following operations: Library: NumPy

- a) Create multi-dimensional arrays and find its shape and dimension
- b) Create a matrix full of zeros and ones
- c) Reshape and flatten data in the array
- d) Append data vertically and horizontally
- e) Apply indexing and slicing on array
- f) Use statistical functions on array - Min, Max, Mean, Median and Standard Deviation

PROCEDURE:

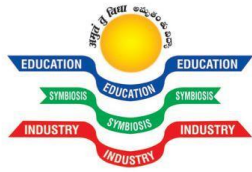
1. Create: Open a new file in Python shell, write a program and save the program with .py extension.
2. Execute: Go to Run -> Run module (F5)

a) Create multi-dimensional arrays and find its shape and dimension

```
import numpy as np
#creation of multi-dimensional array
a=np.array([[1,2,3],[2,3,4],[3,4,5]])
#shape
b=a.shape
print("shape:")
print(a.shape)
#dimension
c=a.ndim
print("dimensions:")
print(a.ndim)
```

b) Create a matrix full of zeros and ones

```
import numpy as np
#matrix full of zeros
z=np. zeros ((2,2))
print("zeros:")
```



```
print(z)
```

#matrix full of ones

```
o=np. ones ((2,2))
```

```
print("ones:")
```

```
print(o)
```

c) Reshape and flatten data in the

```
import numpy as np
```

```
a=np.array([[1,2,3,4],[2,3,4,5],[3,4,5,6],[4,5,6,7]])
```

```
b=a.reshape(4,2,2)
```

```
print("reshape:")
```

```
print(b)
```

#matrix flatten

```
c=a.flatten()
```

```
print("flatten:")
```

```
print(c)
```

d) Append data vertically and horizontally

```
import numpy as np
```

#Appending data vertically

```
x=np.array([[10,20],[80,90]])
```

```
y=np.array([[30,40],[60,70]])
```

```
v=np.vstack((x,y))
```

```
print("vertically:")
```

```
print(v)
```

#Appending data horizontally

```
h=np.hstack((x,y))
```

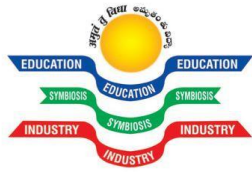
```
print("horizontally:")
```

```
print(h)
```

e) Apply indexing and slicing on array #indexing

```
import numpy as np
```

```
a=np.array([[1,2,3,4],[2,3,4,5],[3,4,5,6],[4,5,6,7]])
```



```
temp = a[[0, 1, 2, 3],[1, 1, 1, 1]]
print('indexing:')
print(temp)
#slicing
i=a[:,::2]
print('slicing:')
print(i)
```

f) Use statistical functions on array - Min, Max, Mean, Median and Standard Deviation

```
import numpy as np
#min for finding minimum of an array
a=np.array([[1,3,-1,4],[3,-2,1,4]])
b=a.min()
print("minimum:",b)
#max for finding maximum of an array
c=a.max()
print("maximum:",c)
#mean
a=np.array([1,2,3,4,5])
d=a.mean()
print("mean:",d)
#median
e=np.median(a)
print("median:",e)
#standard deviation
f=a.std()
print("standard deviation:",f)
```